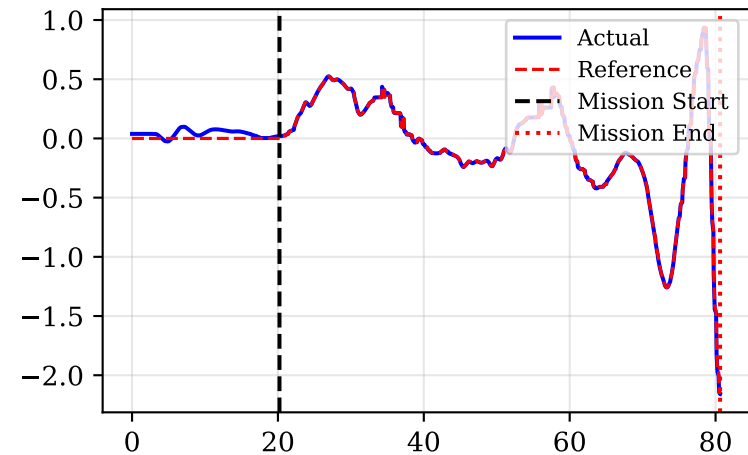
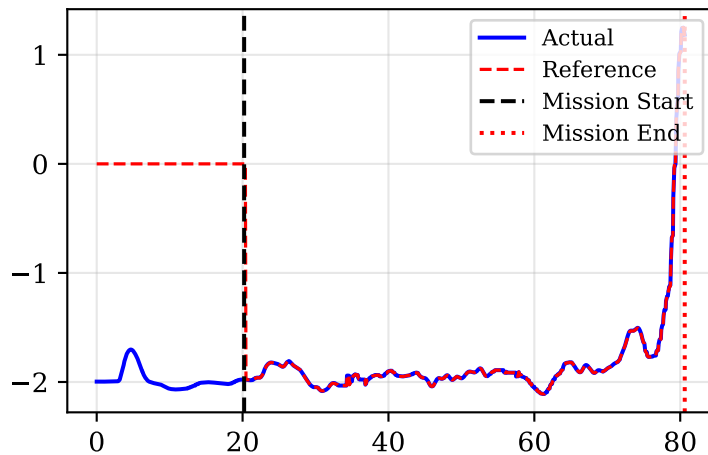


Drone Position vs MPC optimal trajectory

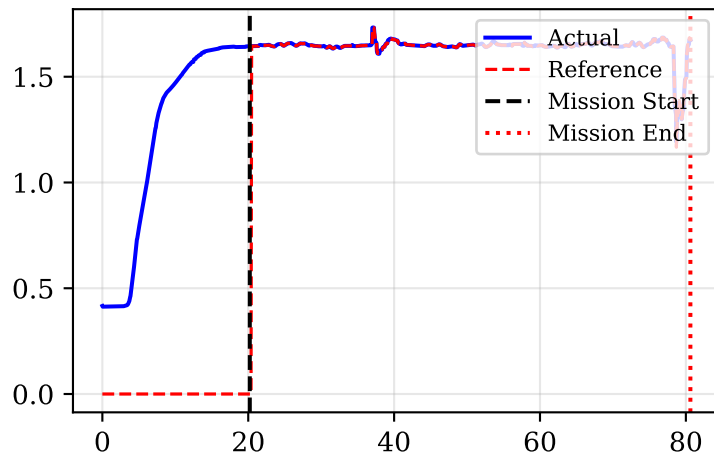
Position X [m]



Position Y [m]

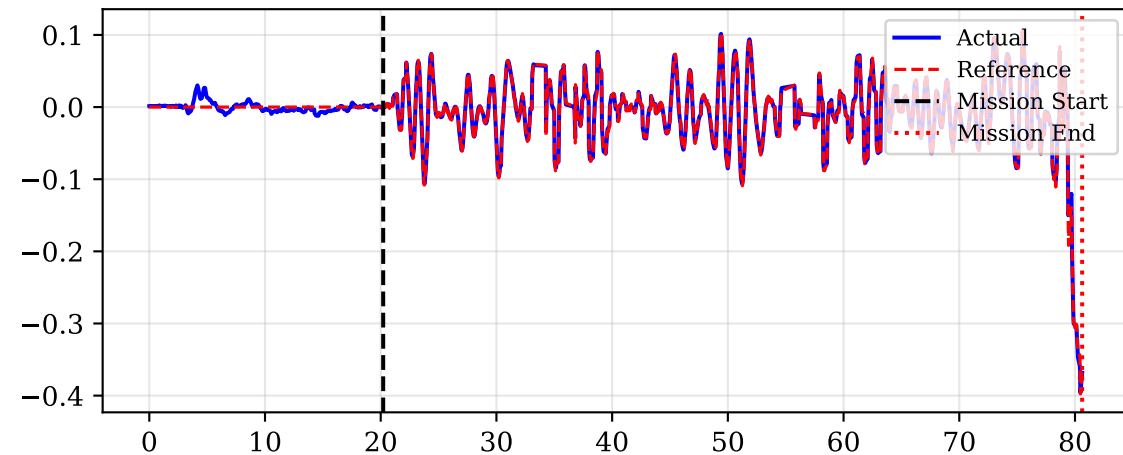


Position Z [m]

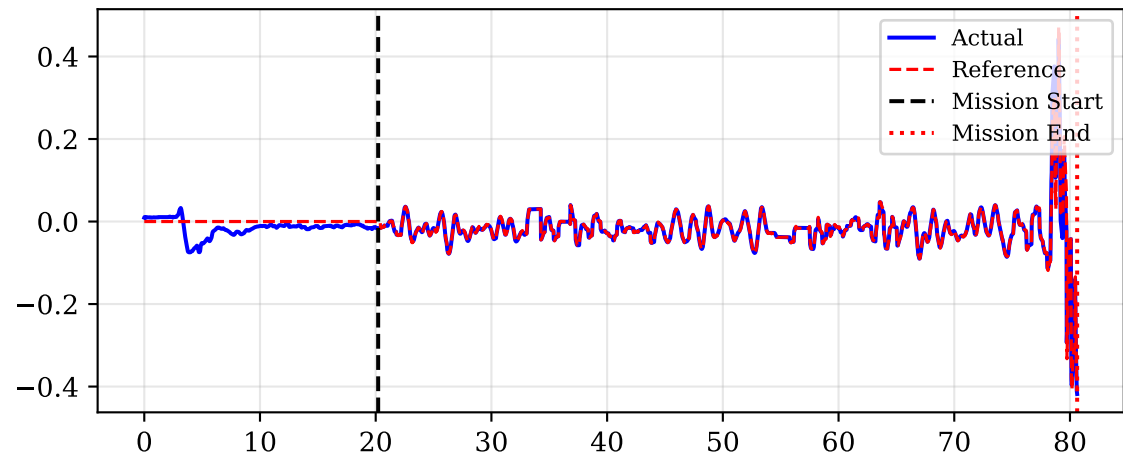


Drone Orientation (Roll/Pitch)

Roll [rad]

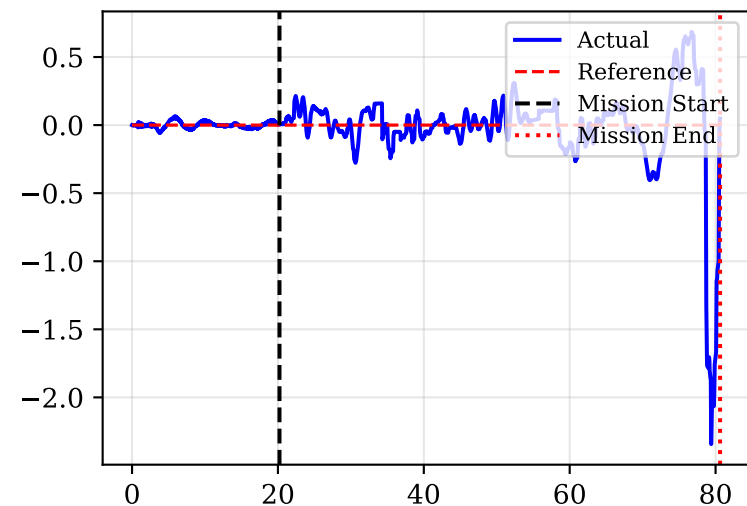


Pitch [rad]

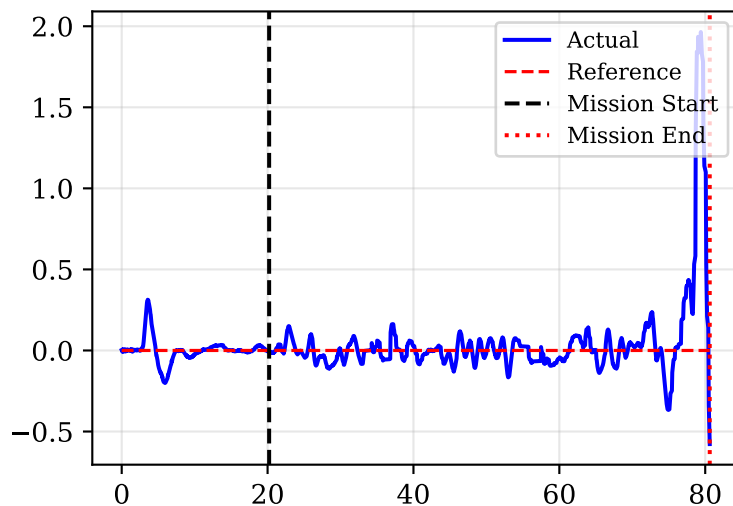


Drone Velocities vs MPC Reference

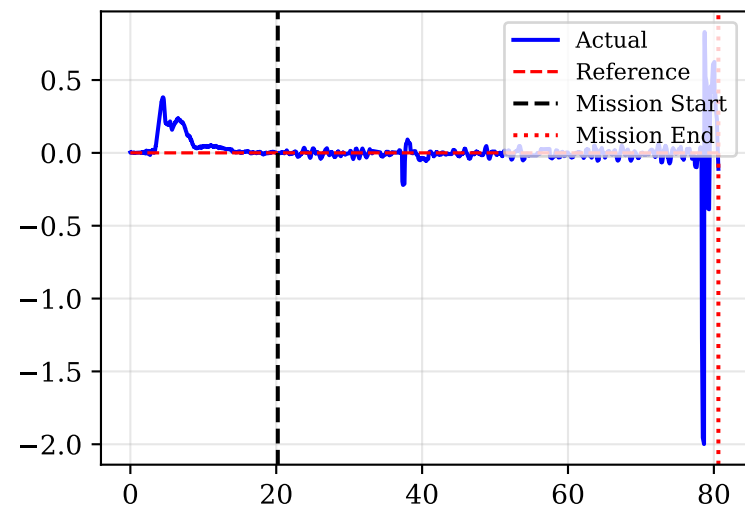
Vel X [m/s]



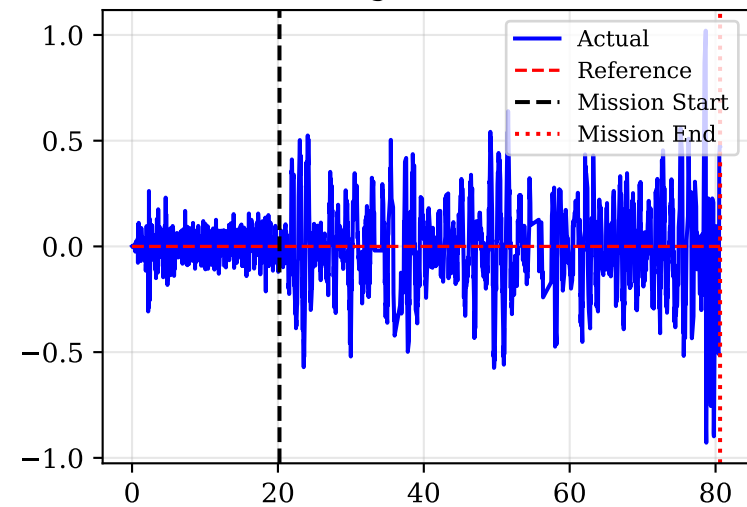
Vel Y [m/s]



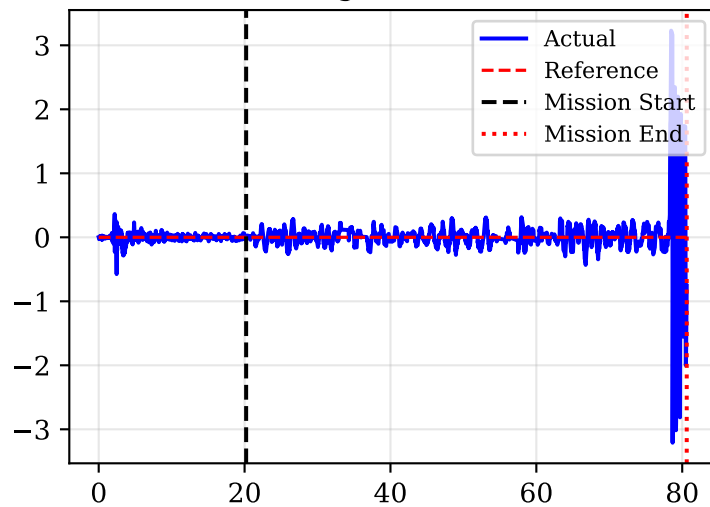
Vel Z [m/s]



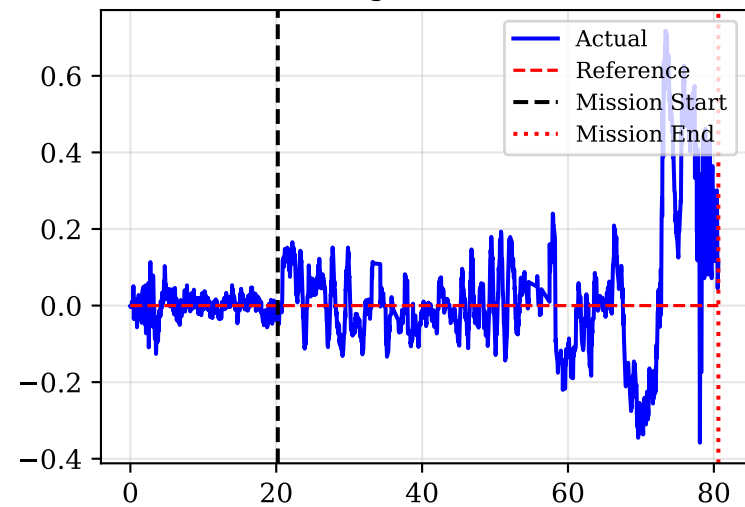
Omega X [rad/s]



Omega Y [rad/s]

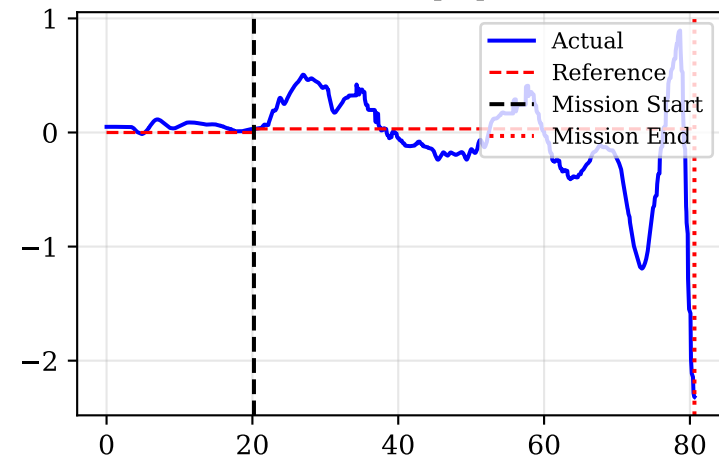


Omega Z [rad/s]

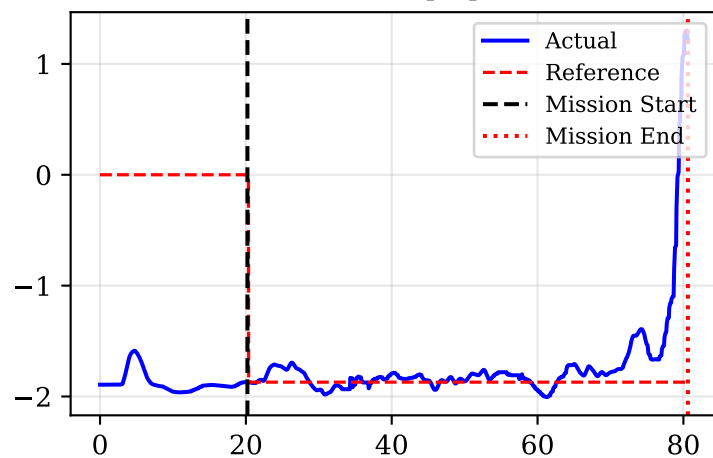


Camera Cartesian Position vs Derived Target

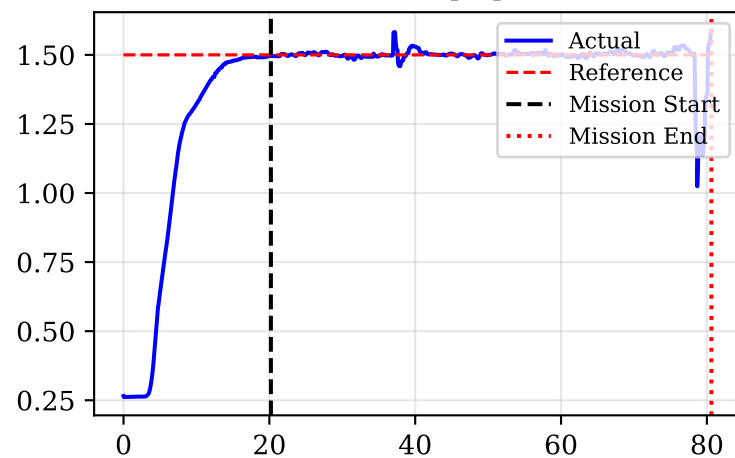
Cam X [m]



Cam Y [m]

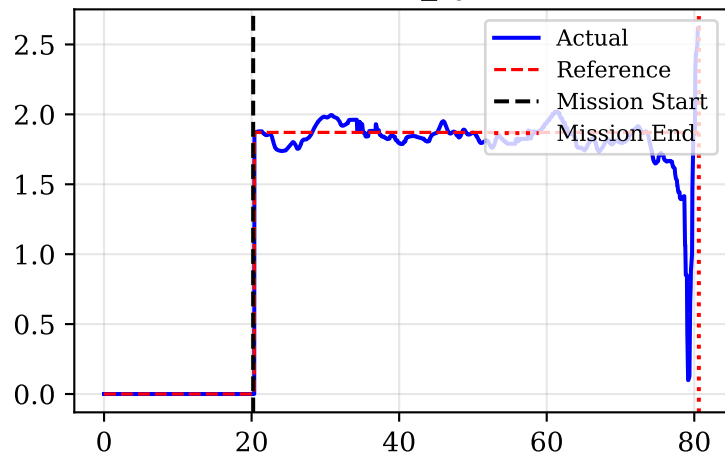


Cam Z [m]

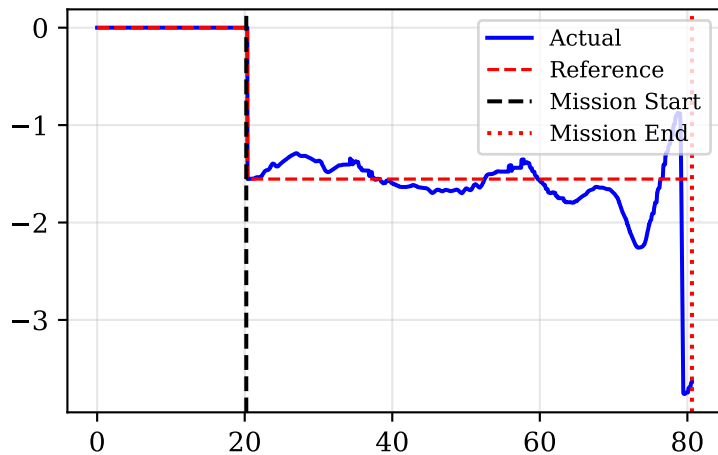


Cylindrical PoV Tracking (World Frame)

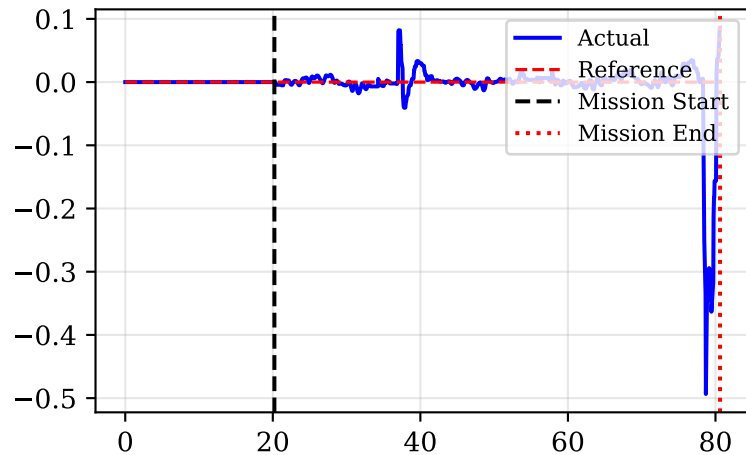
Distance r_{cyl} [m]



Azimuth β [rad]

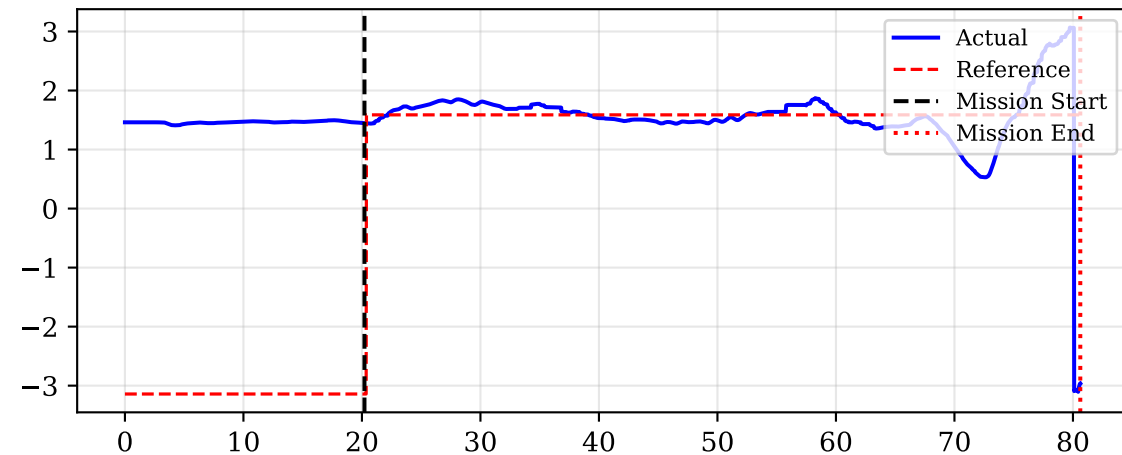


Elevation z [m]

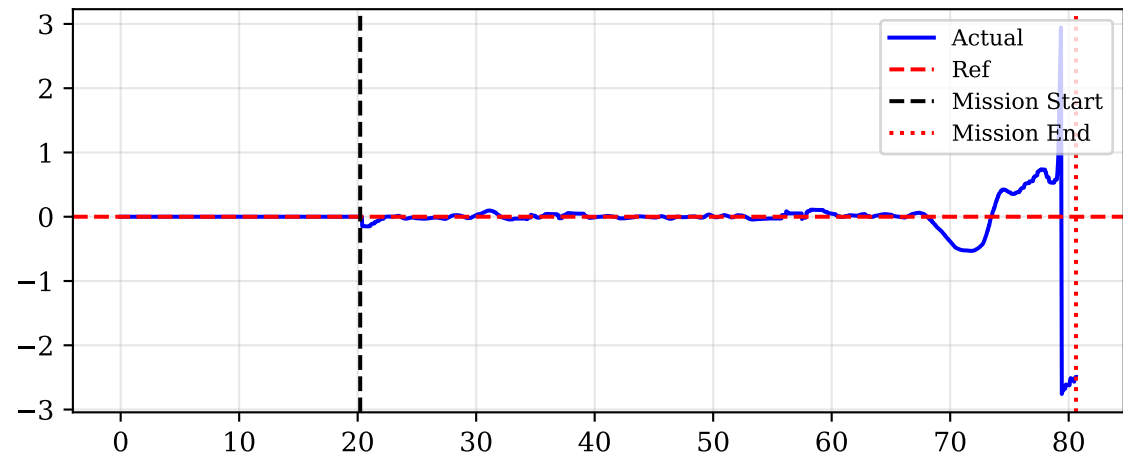


Yaw Tracking: Drone Pointing Toward Target

Yaw Actual vs Desired [rad]

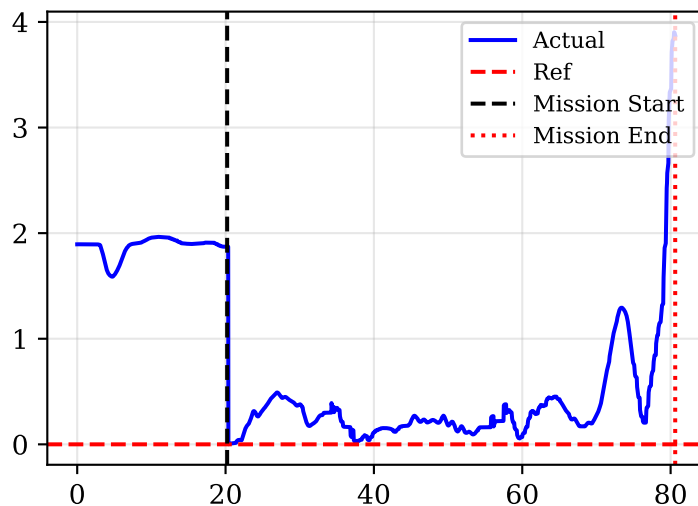


Yaw Error [rad]

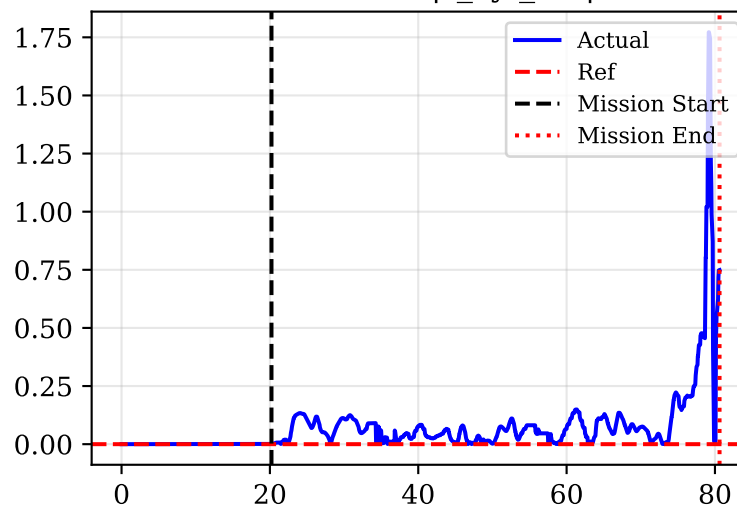


Primary Tracking Errors (Cylindrical)

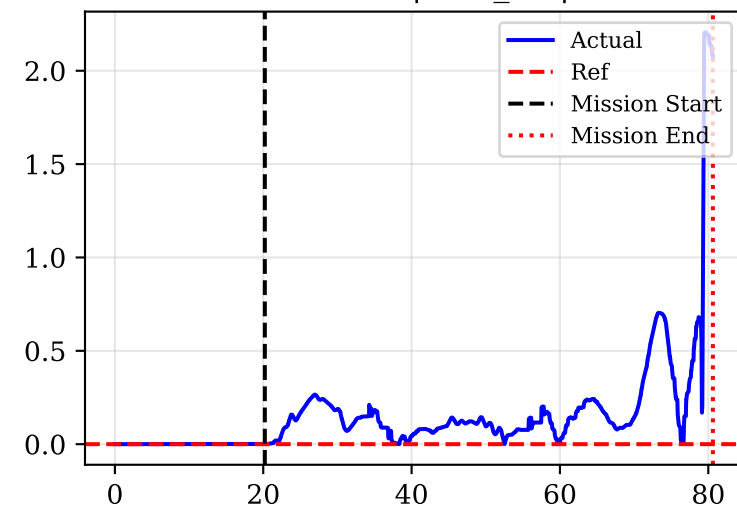
Cam XY Error [m]



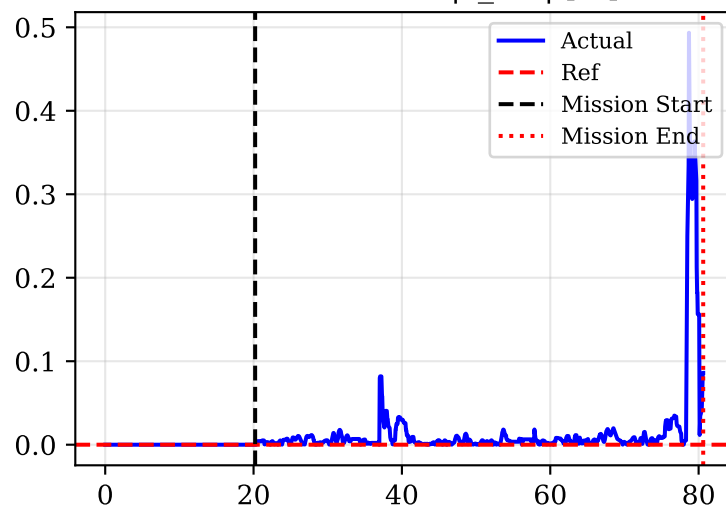
Distance Error $|r_{cyl_err}|$ [m]



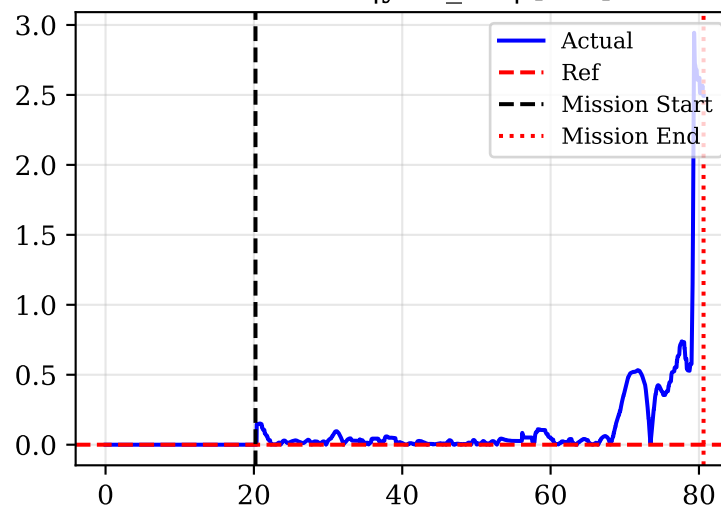
Azimuth Error $|\beta_{err}|$ [rad]



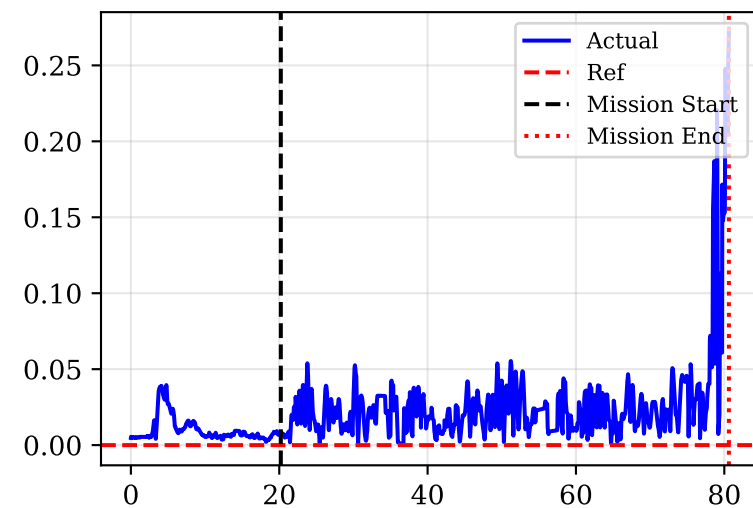
Elevation Error $|z_{err}|$ [m]



Yaw Error $|\text{yaw_err}|$ [rad]

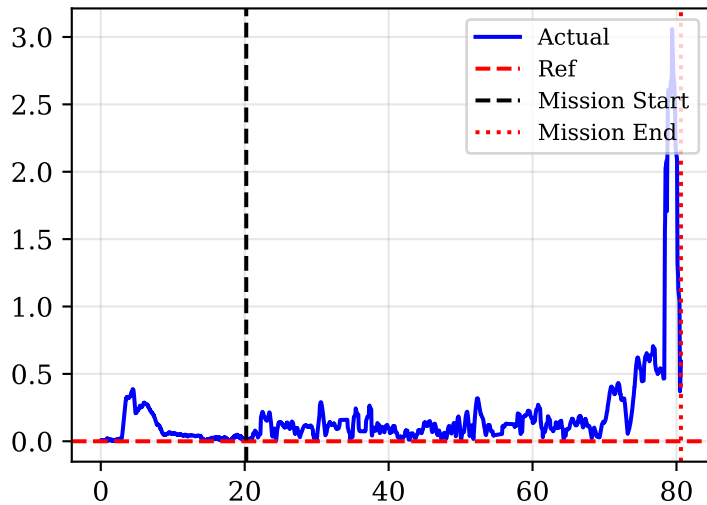


Norm Roll/Pitch Error

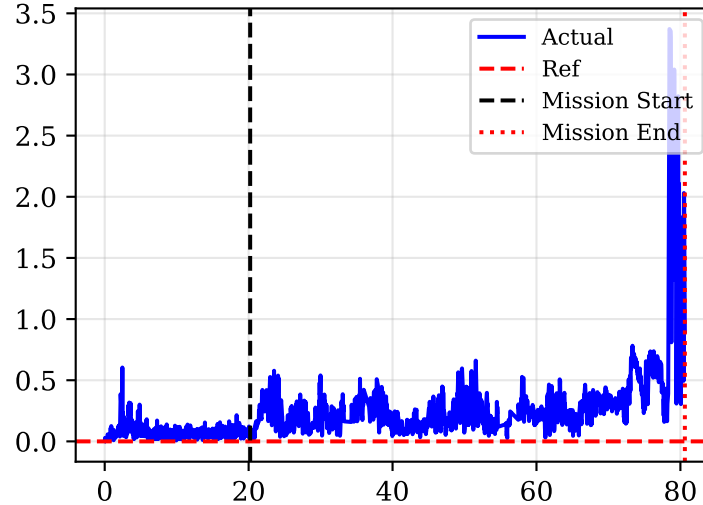


Dynamic States Errors and Feedforward Derivatives

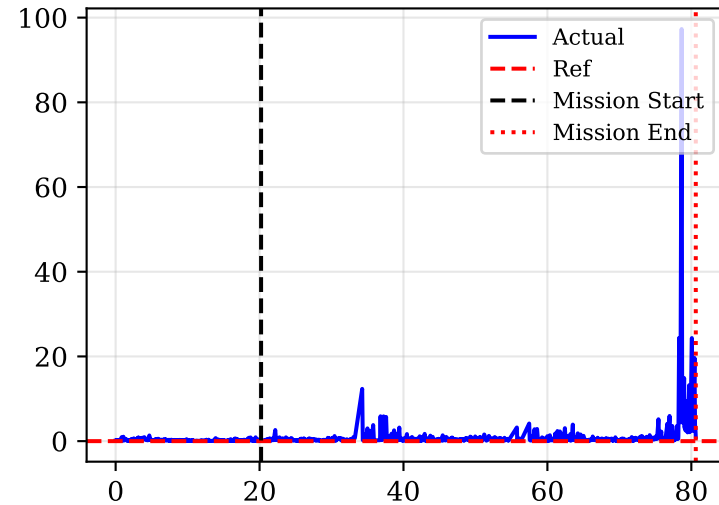
Norm Vel Error [m/s]



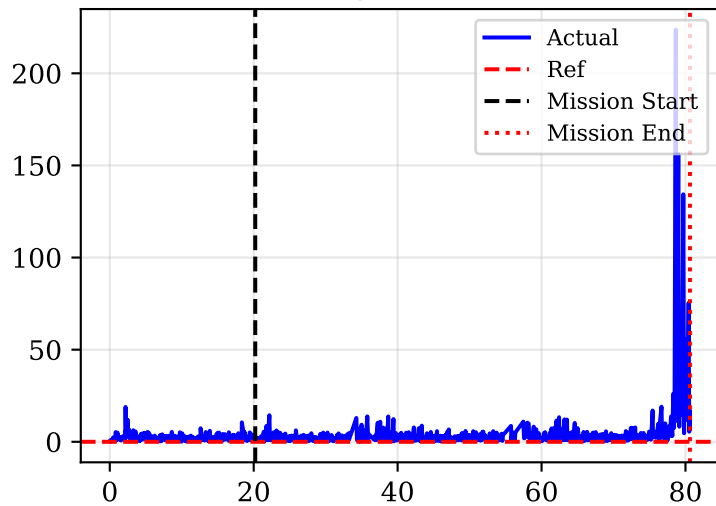
Norm Omega Error [rad/s]



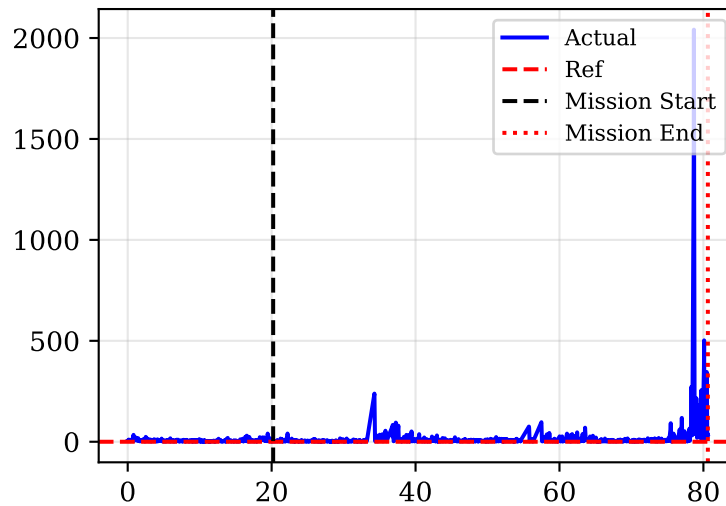
Norm Acc [m/s^2]



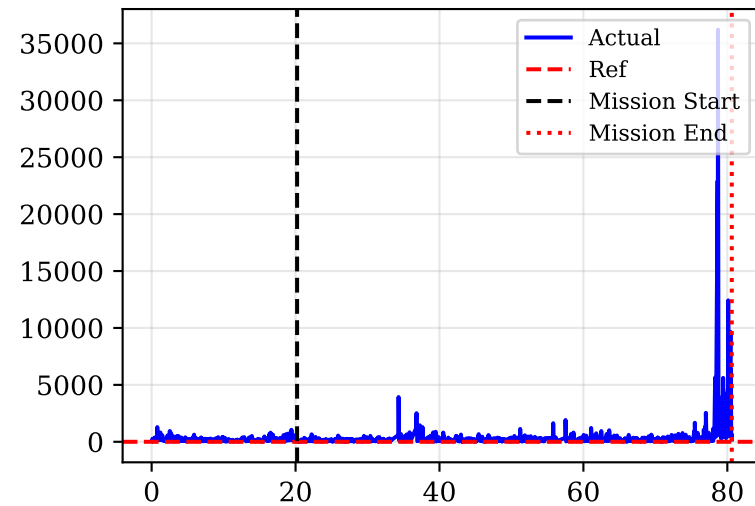
Norm AngAcc [rad/s^2]



Norm Jerk [m/s^3]

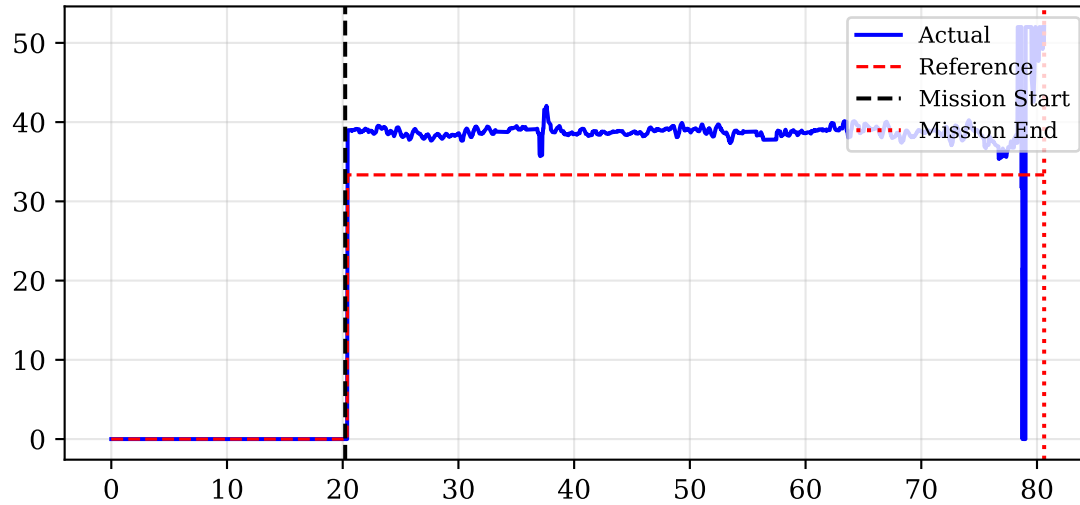


Norm Snap [m/s^4]

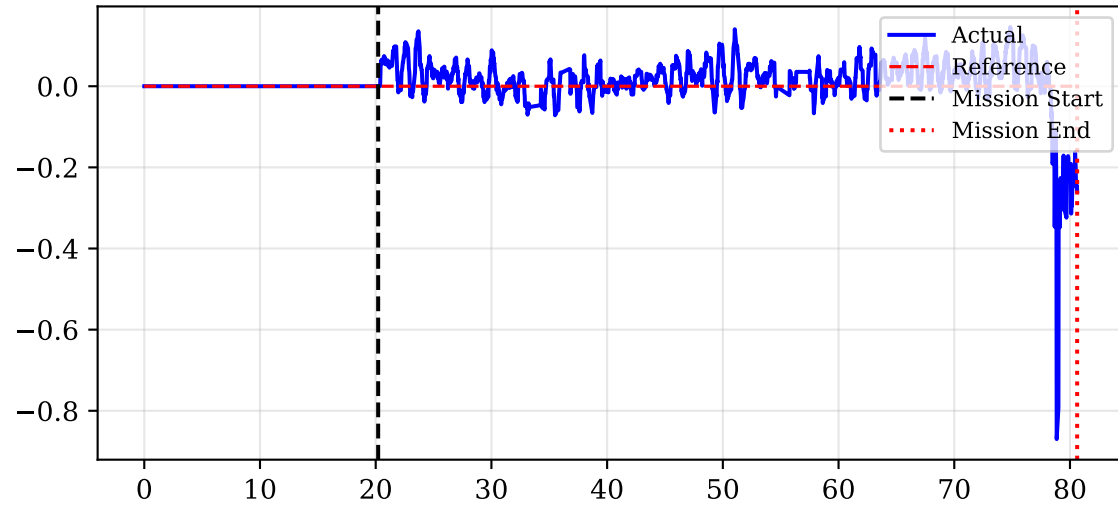


Control Wrench (Hover Force = 20.24N)

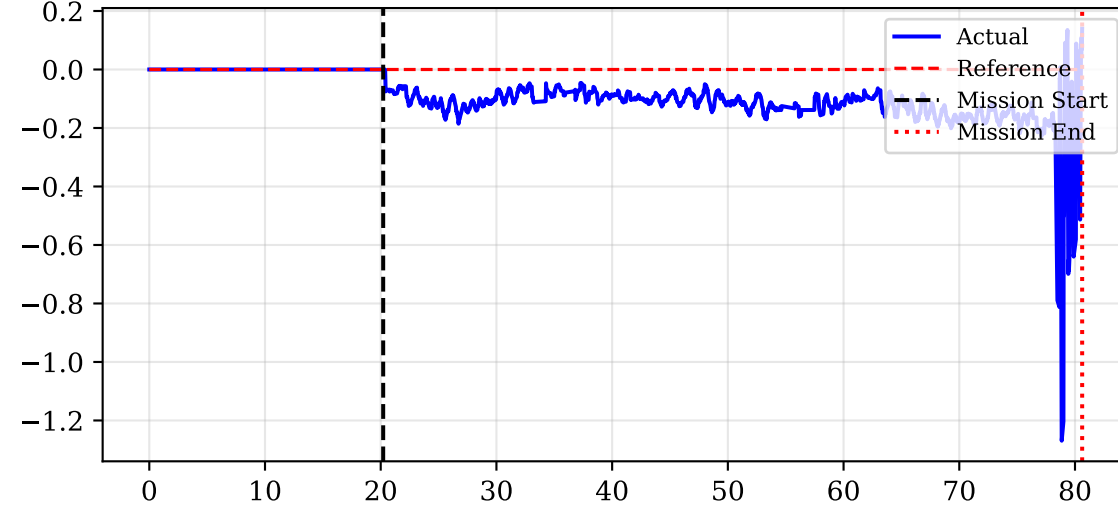
Force Z (Thrust) [N]



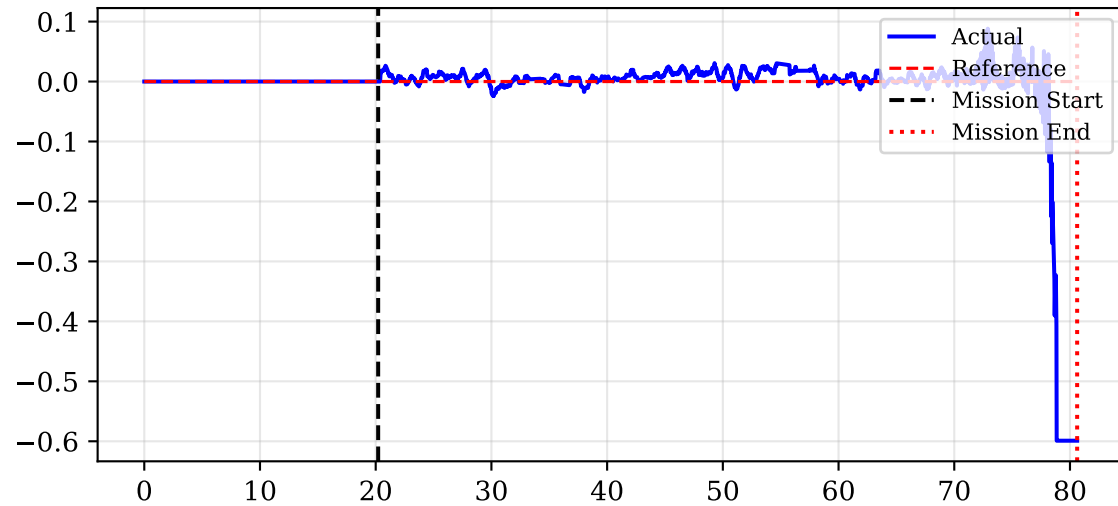
Torque X [Nm]



Torque Y [Nm]

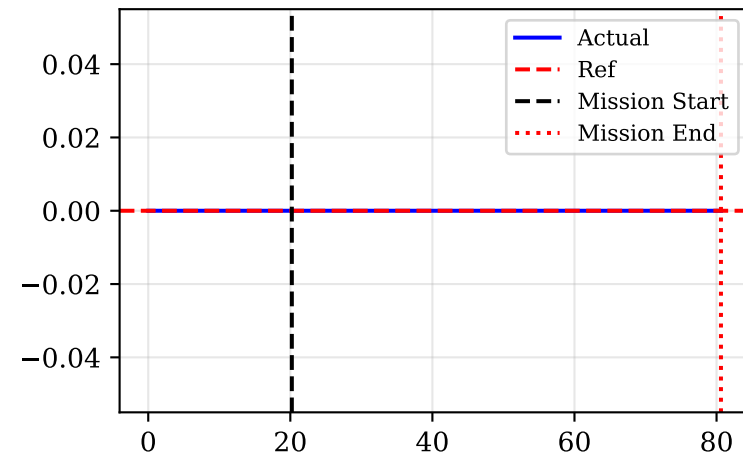


Torque Z [Nm]

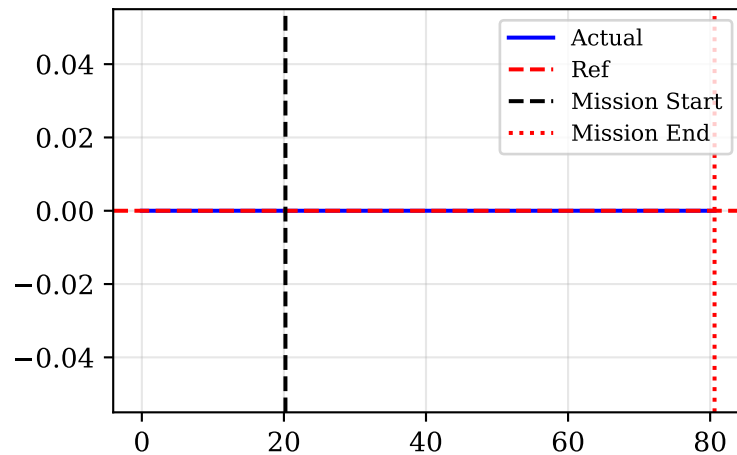


Haptic Feedback Forces Transmitted to haptic device

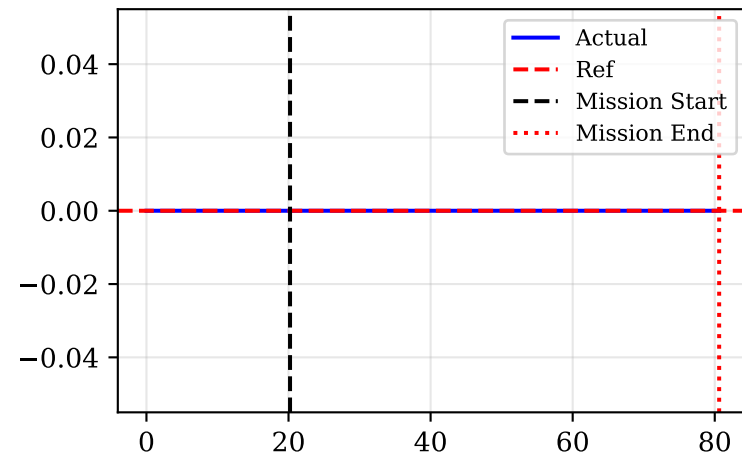
Force X (Zoom) [N]



Force Y (Pan) [N]

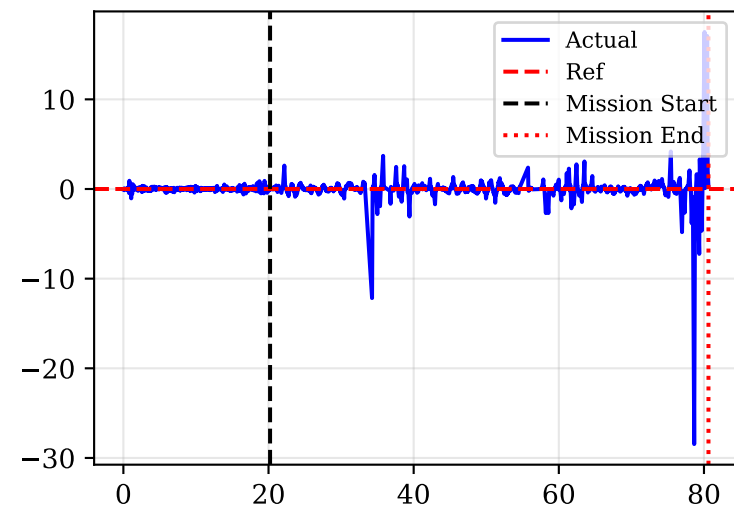


Force Z (Altitude) [N]

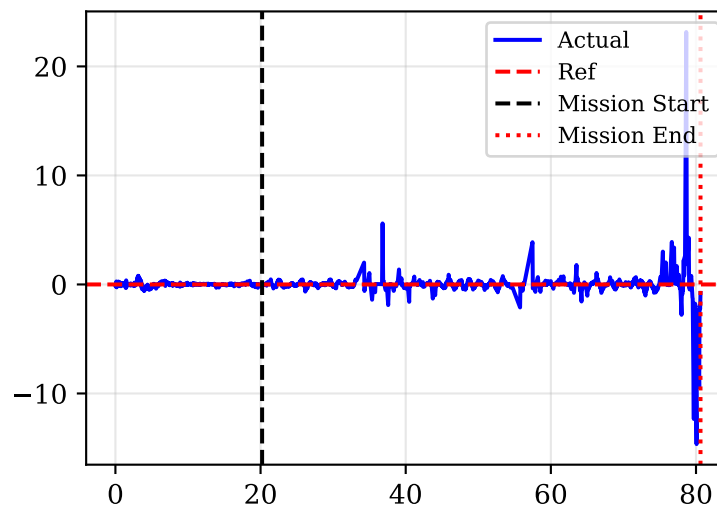


Drone Linear and Angular Accelerations

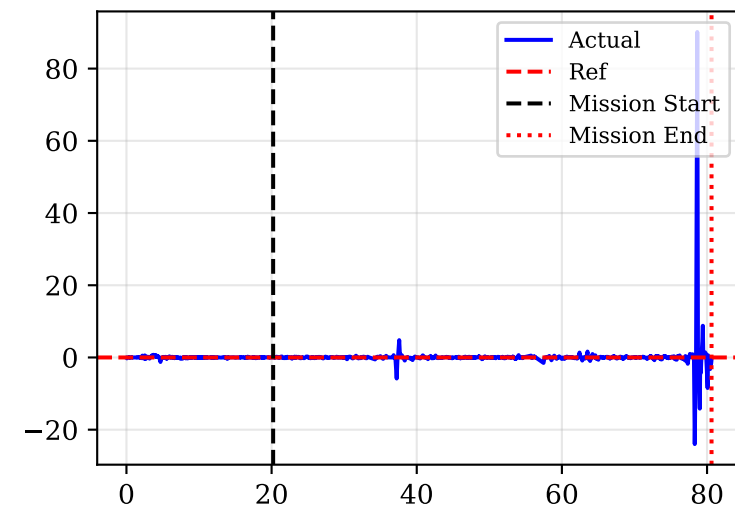
Linear Acc X [m/s²]



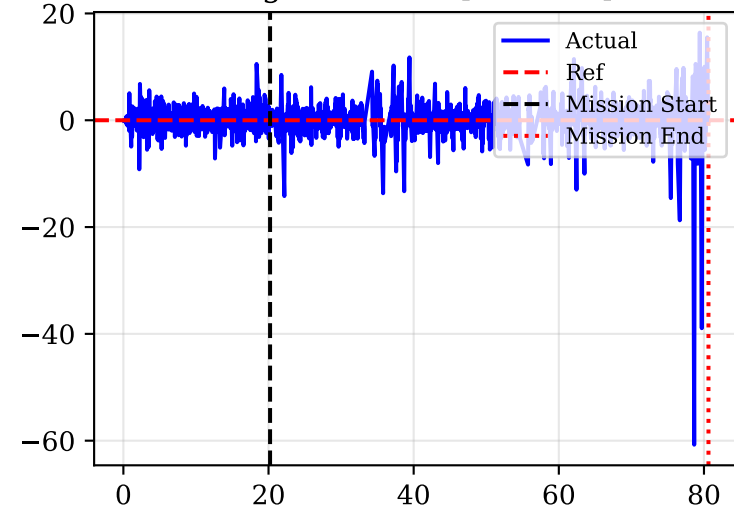
Linear Acc Y [m/s²]



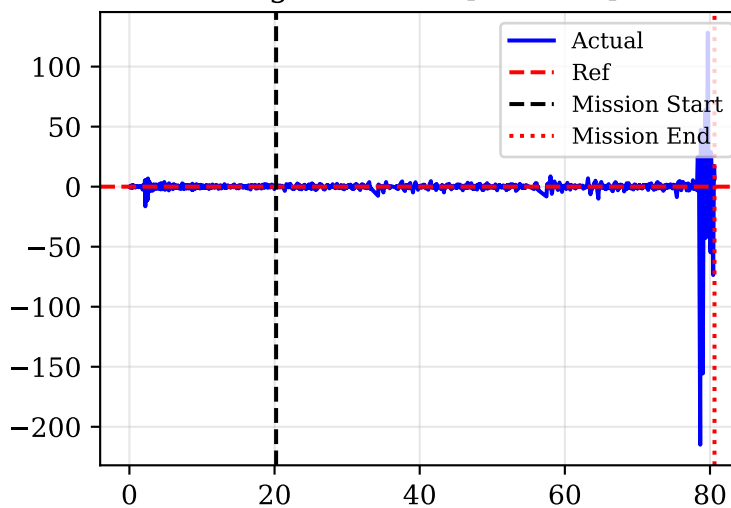
Linear Acc Z [m/s²]



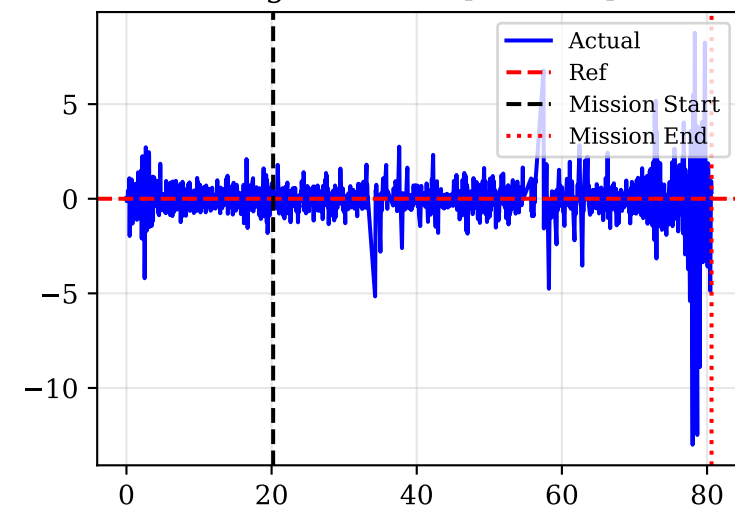
Angular Acc X [rad/s²]



Angular Acc Y [rad/s²]

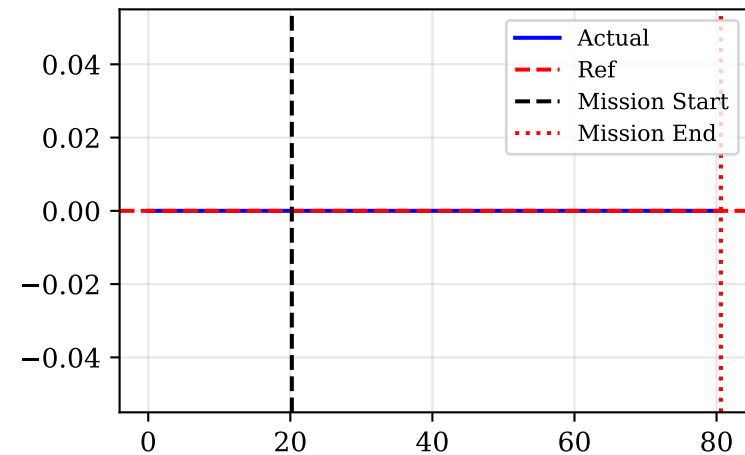


Angular Acc Z [rad/s²]

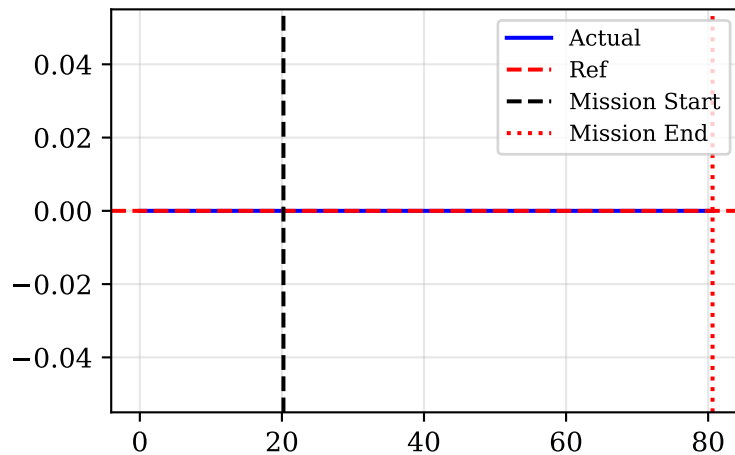


Peg External Contact Forces (FT Sensor)

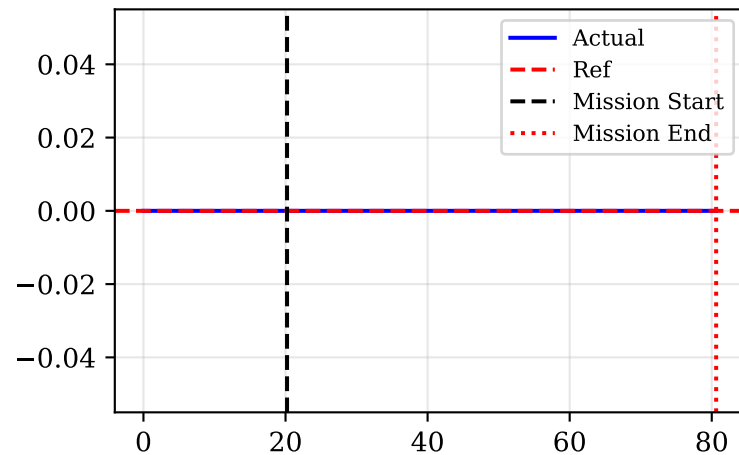
Force X (Sensor) [N]



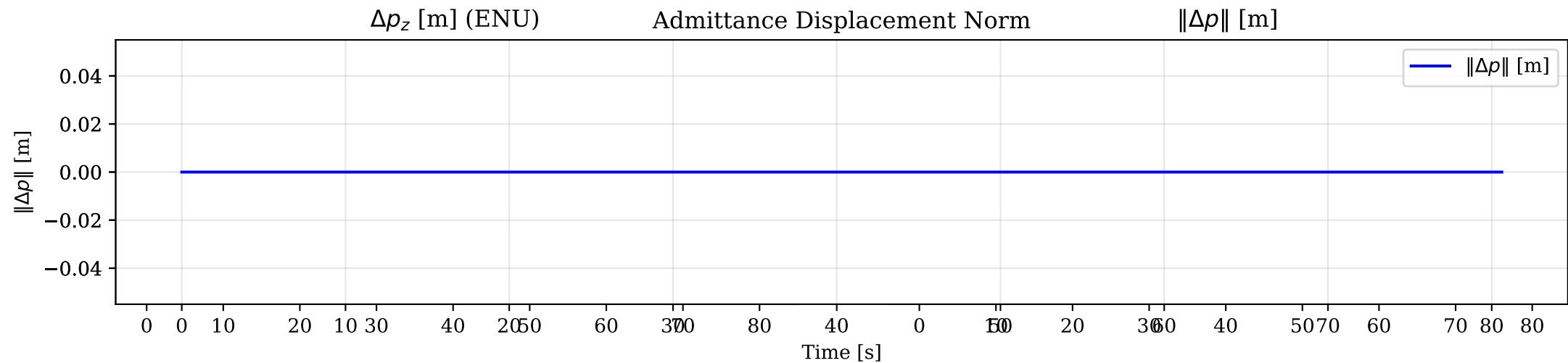
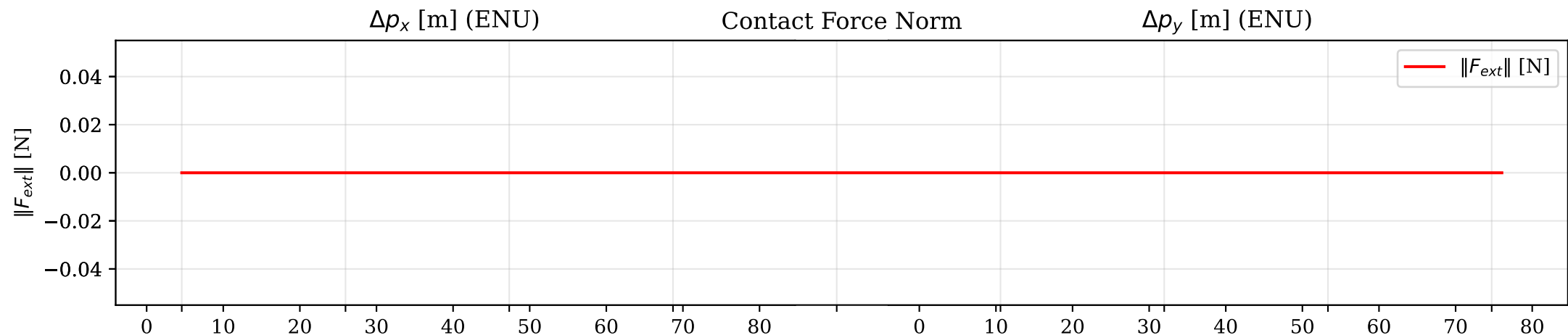
Force Y (Sensor) [N]



Force Z (Sensor) [N]

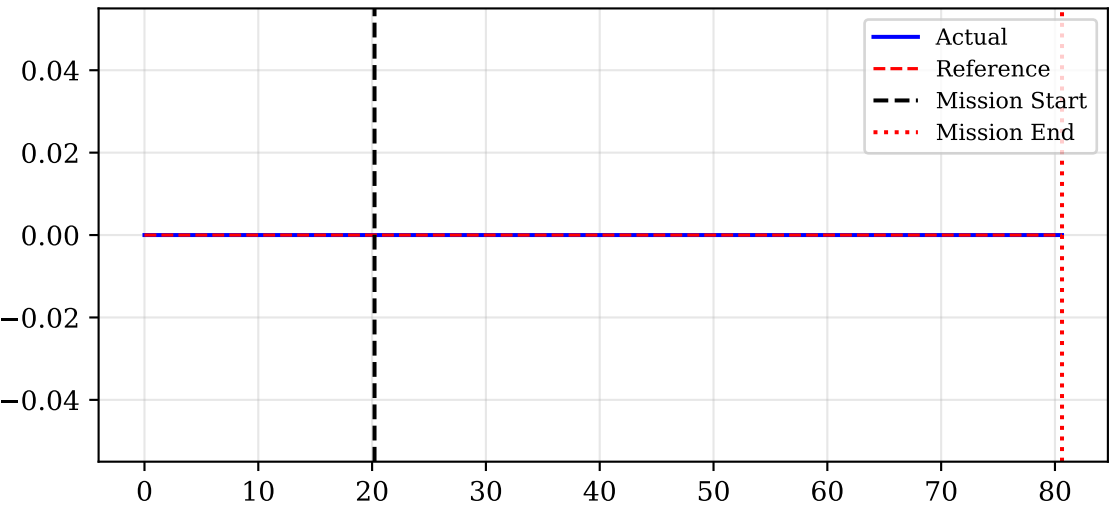


Admittance Effect: Contact Force vs Position Deviation

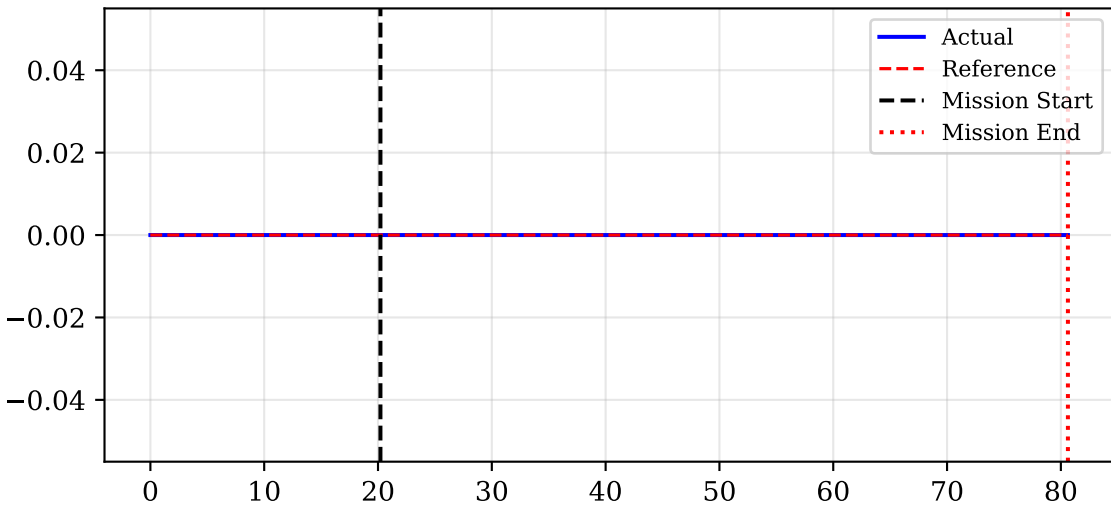


Interaction Drone Position ENU (Actual vs Planner Reference)

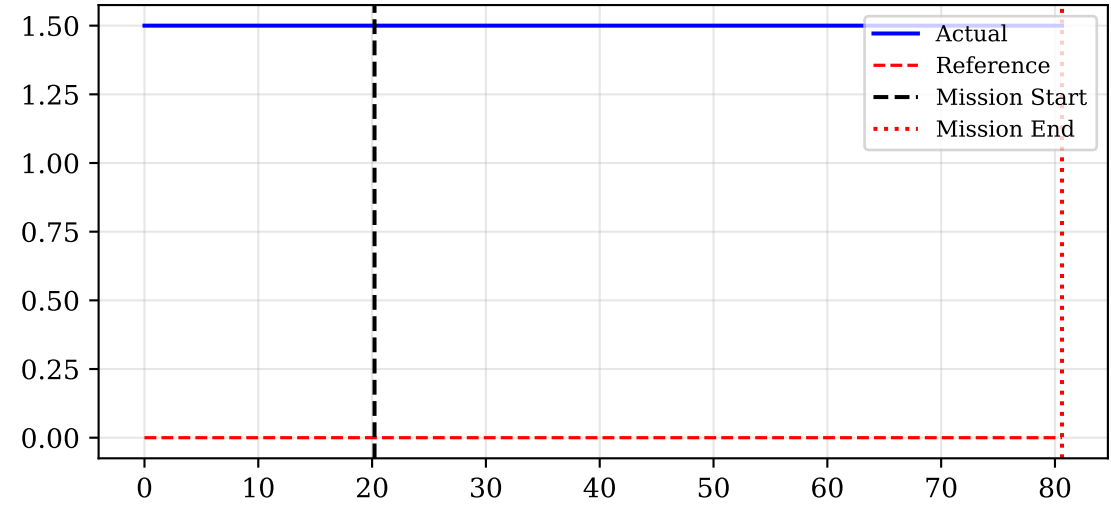
Peg X [m]



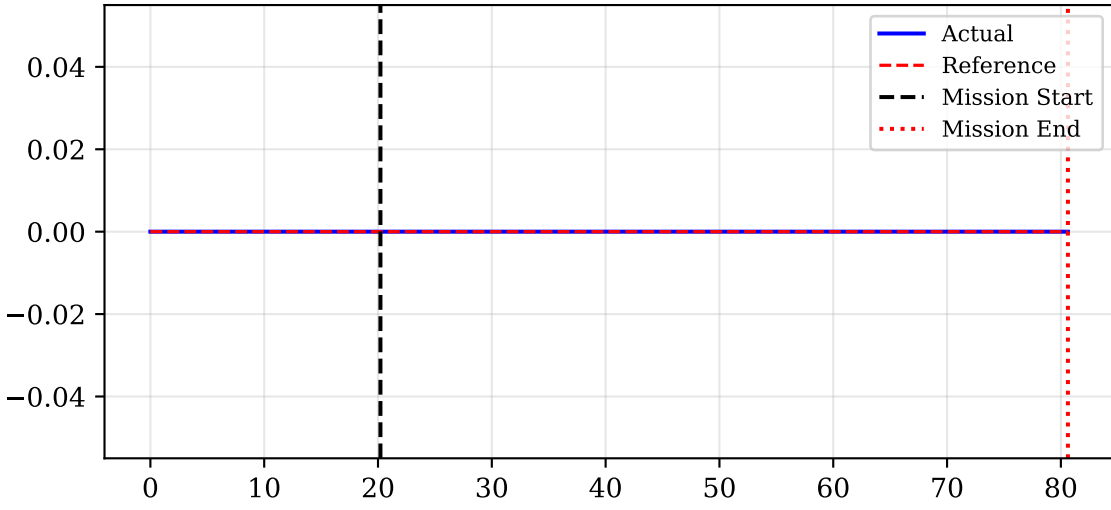
Peg Y [m]



Peg Z [m]

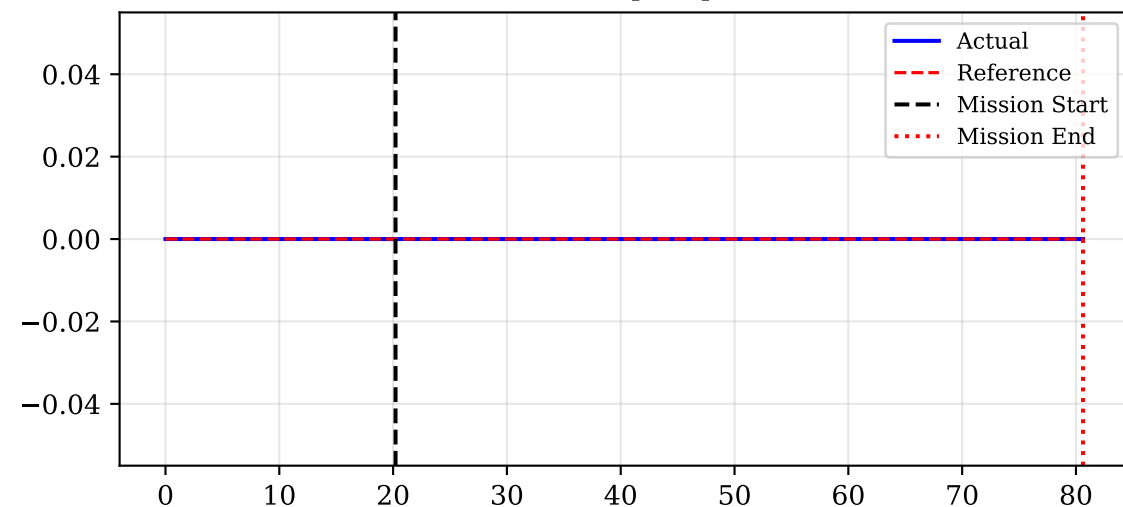


Peg Yaw [rad]

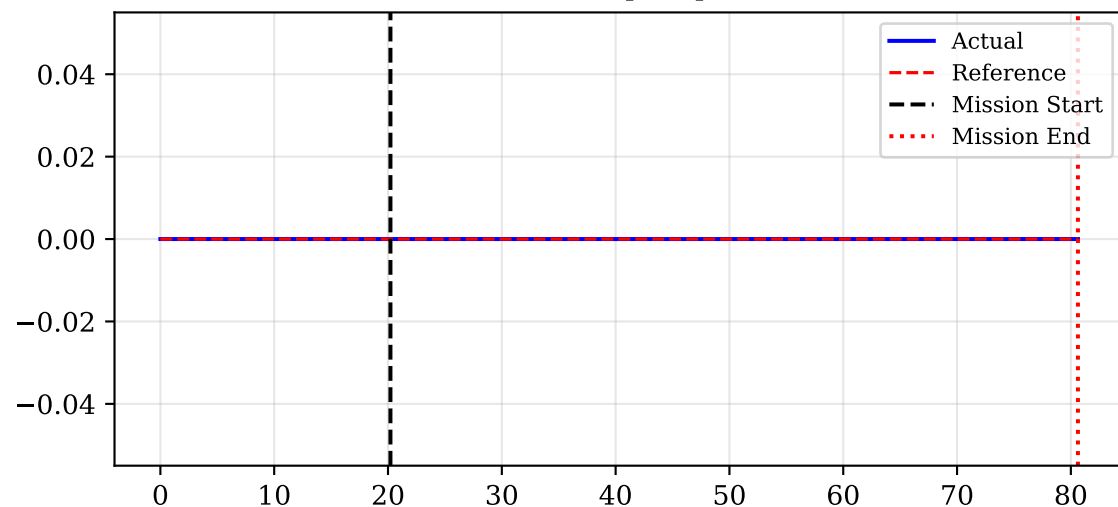


Interaction Drone Velocities (ENU) and Yaw Rate

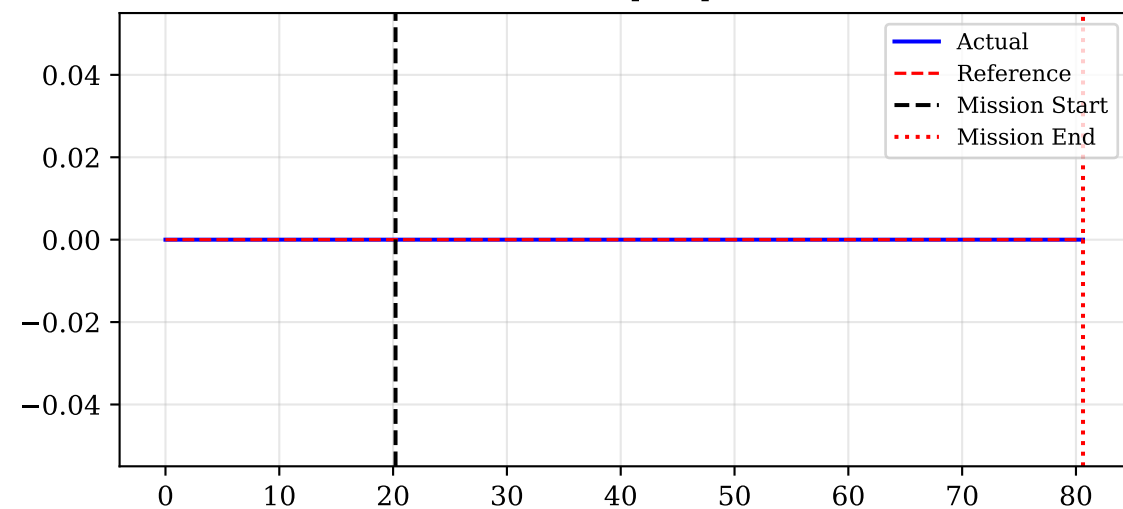
Vel X [m/s]



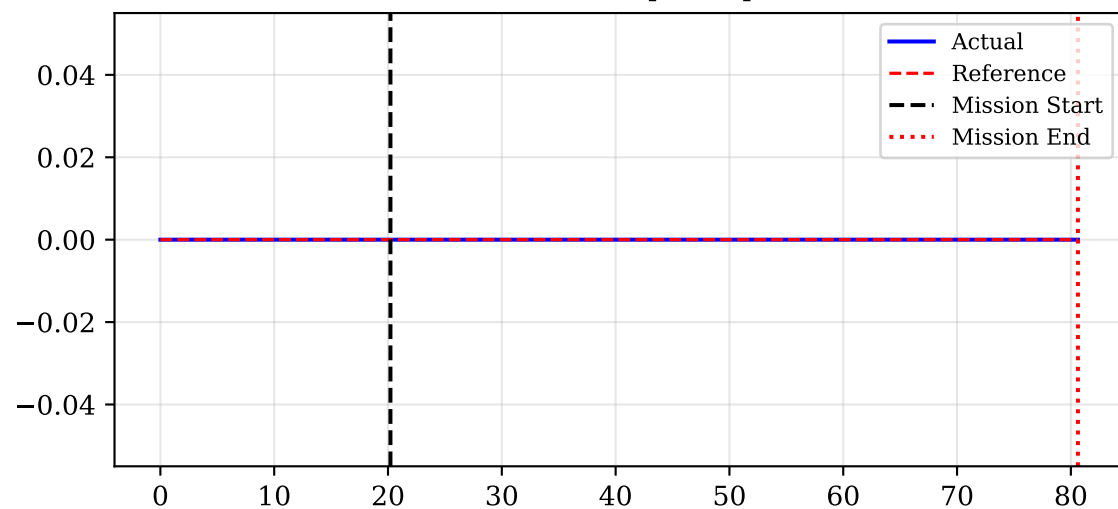
Vel Y [m/s]



Vel Z [m/s]

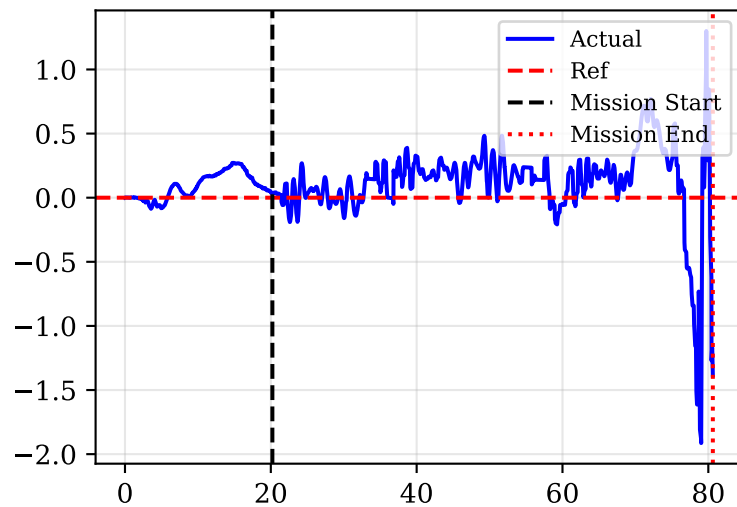


Yaw Rate [rad/s]

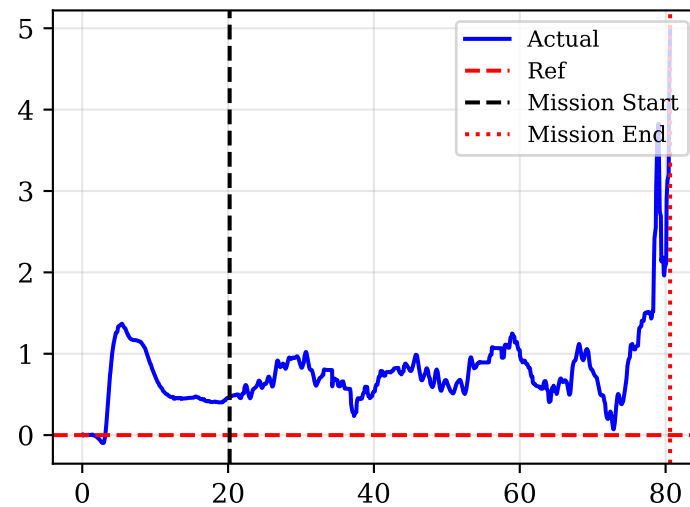


Estimated Wrench (Momentum-Based Estimator)

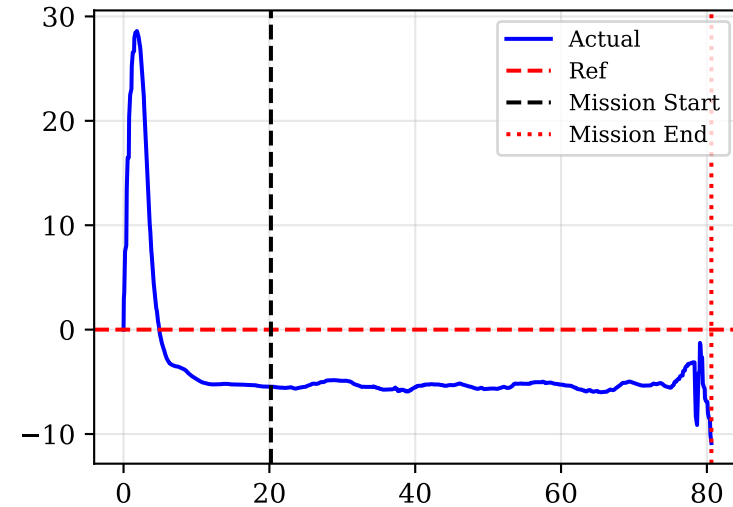
Force X [N]



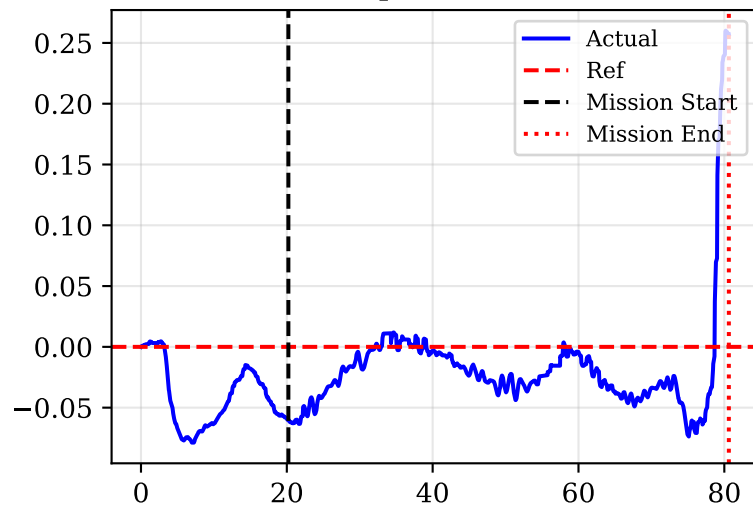
Force Y [N]



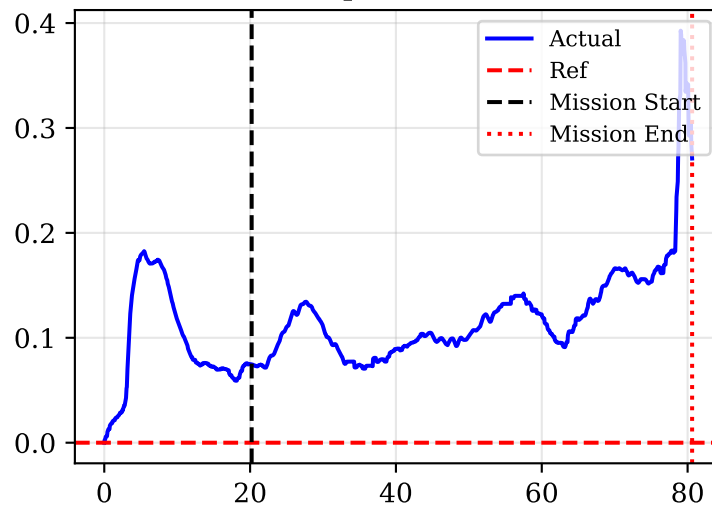
Force Z [N]



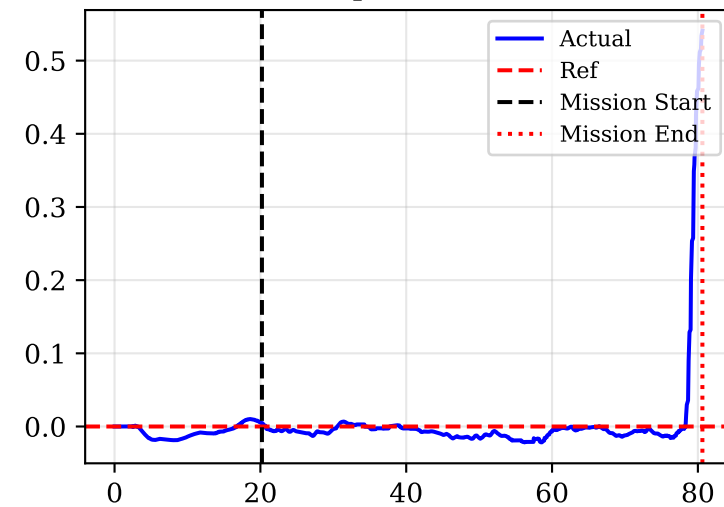
Torque X [Nm]



Torque Y [Nm]

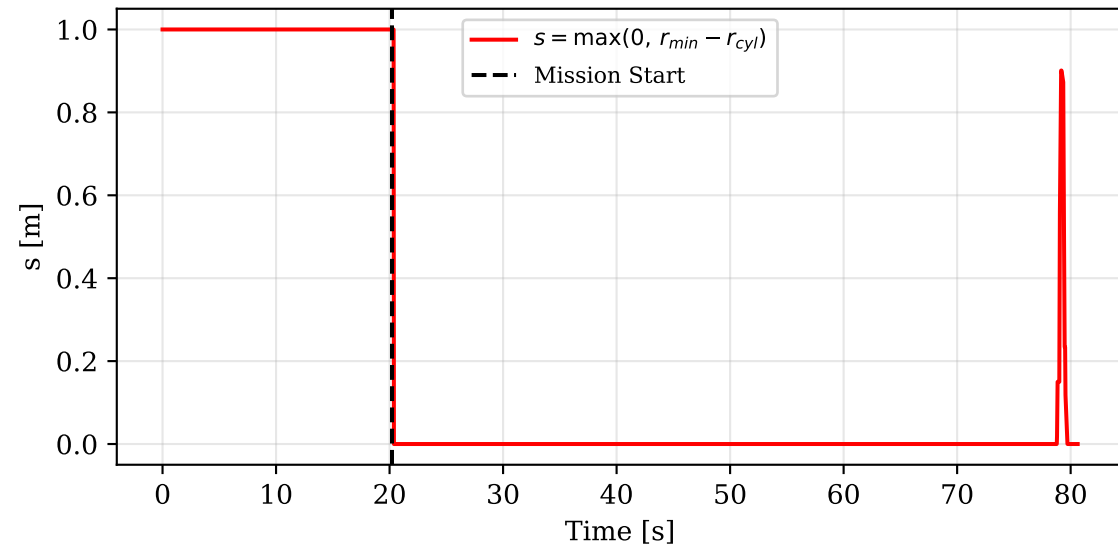


Torque Z [Nm]

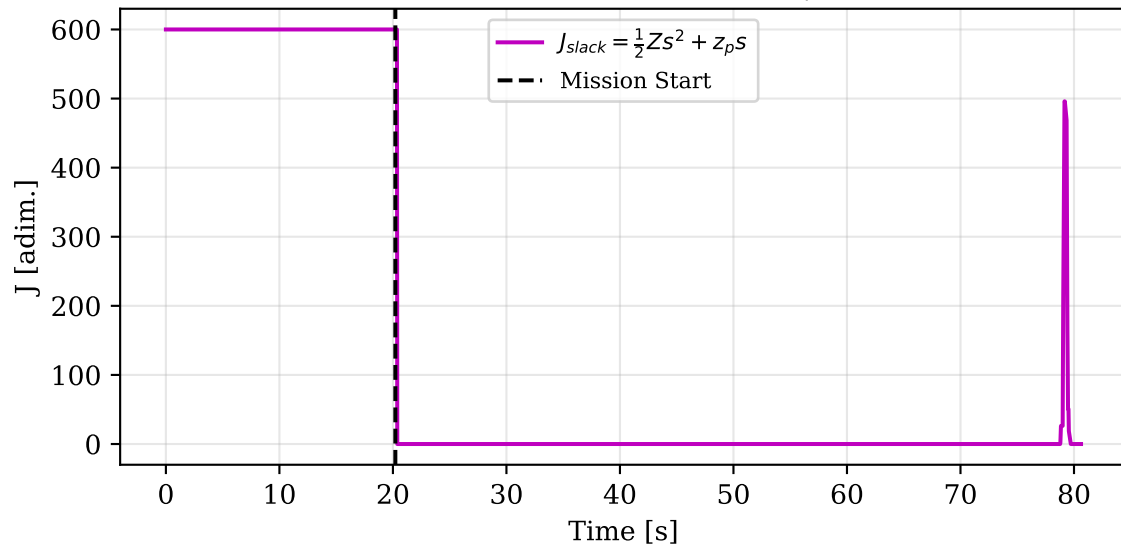


Soft Constraint — Distanza Minima da Oggetto

Violazione geometrica $s(t)$ [$r_{min}=1.0$ m]

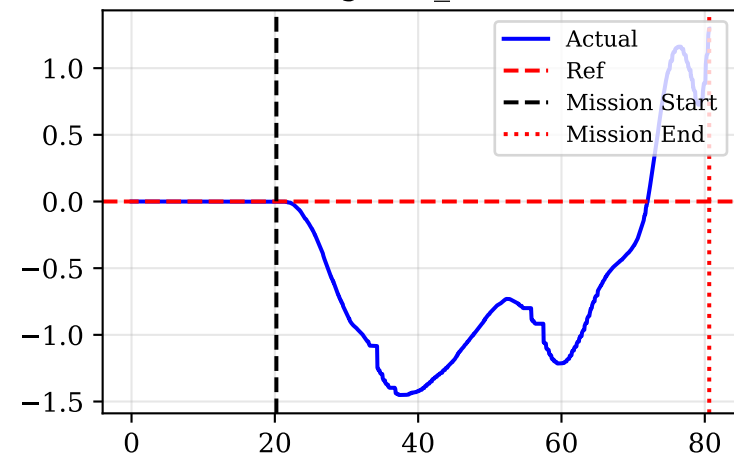


Costo slack $J_{slack}(t)$ [$Z=1e+03$, $z_p=1e+02$]

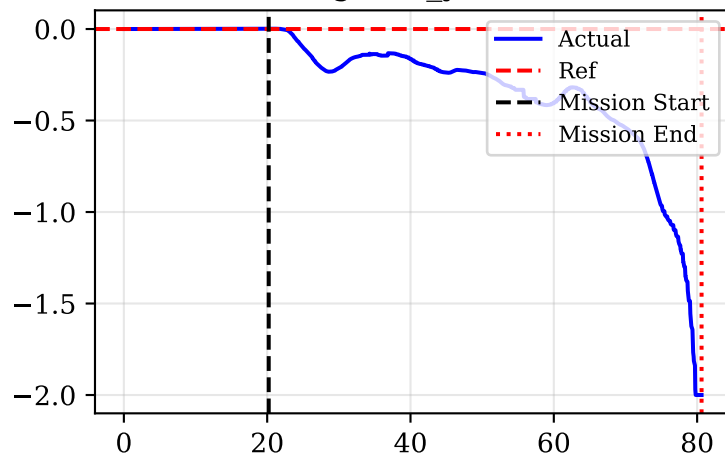


Cartesian Integral Action (Anti-windup clipped)

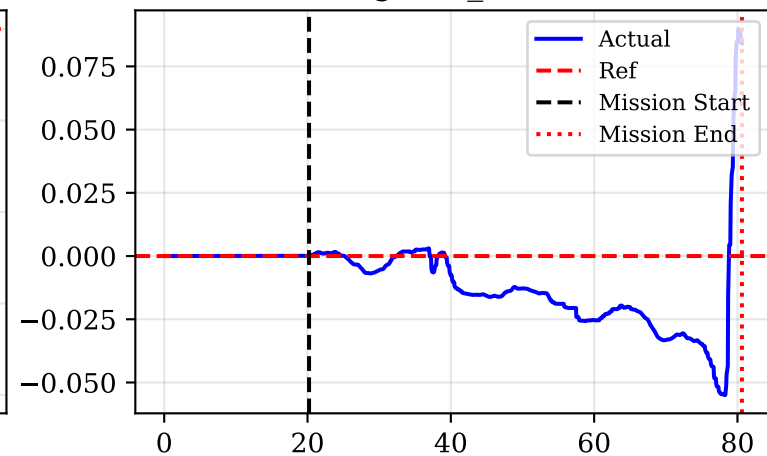
Integral e_x [m*s]



Integral e_y [m*s]

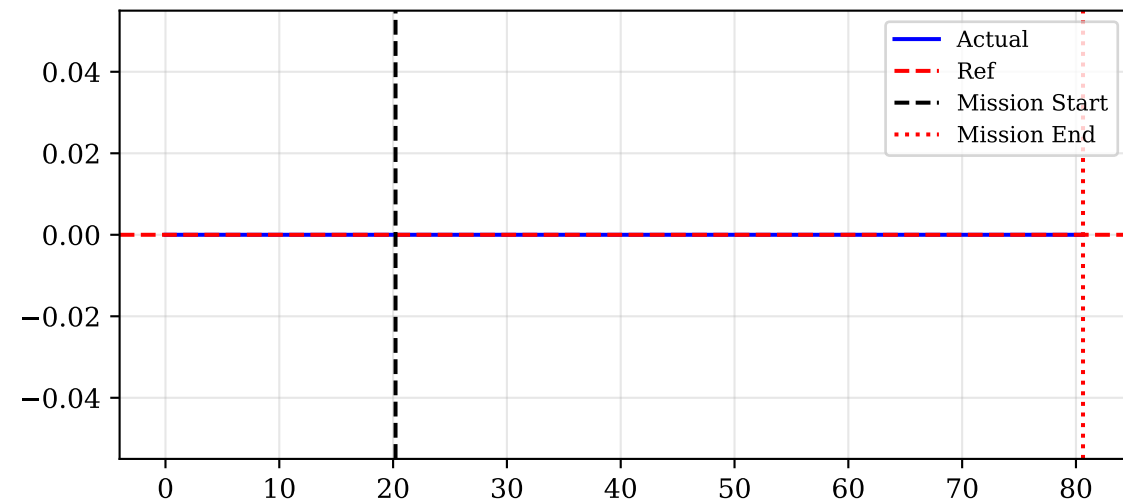


Integral e_z [m*s]

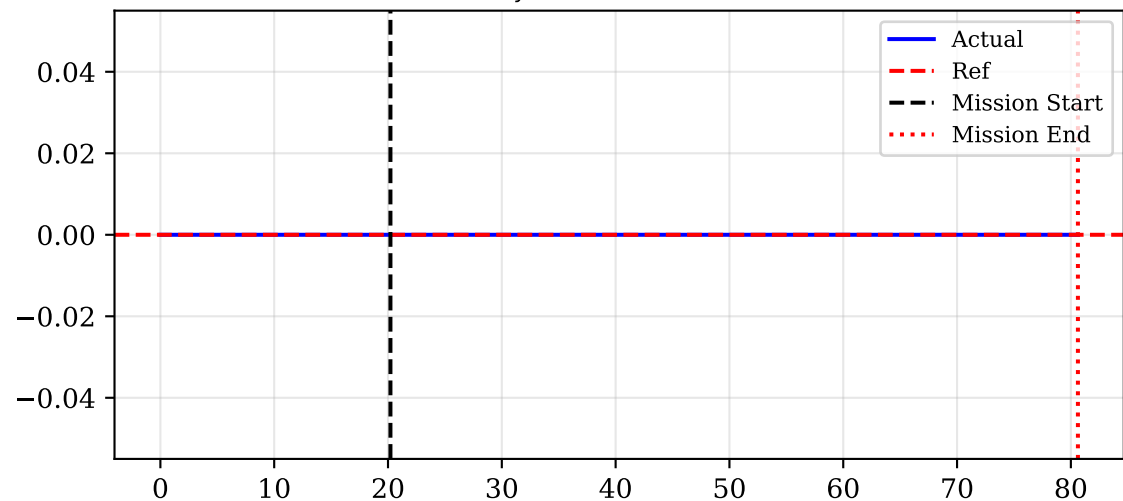


Admittance Displacement in Sensor Frame

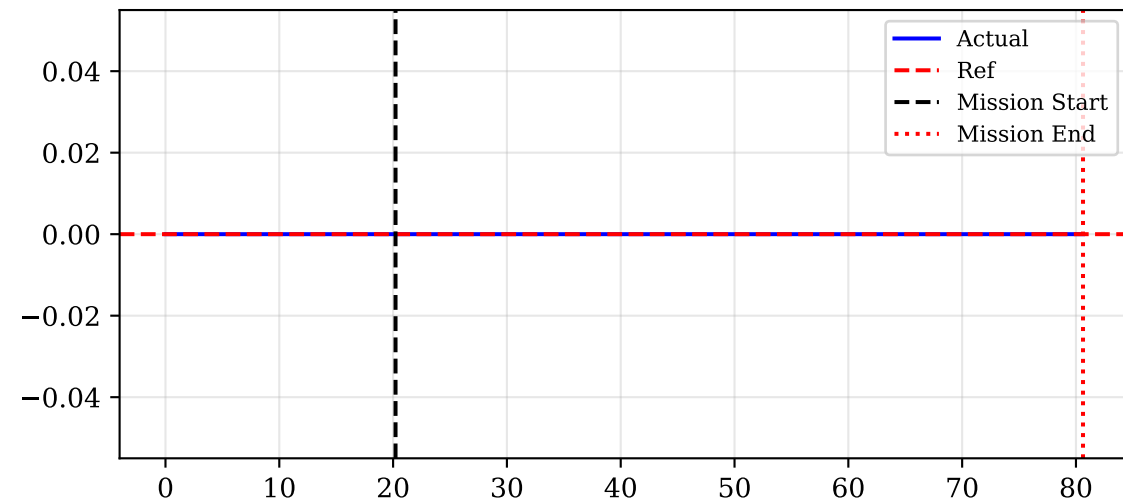
Δp_{sx} [m] (Sensor X)



Δp_{sy} [m] (Sensor Y)



Δp_{sz} [m] (Sensor Z)



$\|\Delta p_s\|$ [m]

